Insurance Policy Management Web API using ASP.NET Core + EF Core

Wat you will Learn:

- Creating DbContext
- Configuring DbContext
- Using SQL Server
- Creating DbSet
- Data Annotations
- Fluent API
- Relationships
- Seed Data
- Primary Keys
- Foreign Keys
- Validation
- Migrations
- Update Database
- CRUD Operations
- DTOs
- Service Layer
- Dependency Injection

Agenda:

1. Domain: Insurance Management System
2. Duration: 5-6 Hrs.
3. Modules:
    a. Customers
    b. Insurance Policies
    c. Policy Types
    d. Claims
4. Database / Tables
    a. Customer: CustomerId, FirstName, LastName, Email, Phone, DateOfBirth, City
    b. PolicyType: PolicyTypeId, PolicyTypeName, Description
    c. Policy: PolicyId, PolicyNumber, PremiumAmount, StartDate, EndDate, CustomerId PolicyTypeId
    d. Claim: ClaimId, ClaimAmount, ClaimDate, Status, PolicyId
5. Project Structure
    a. InsuranceManagementAPI
    b. Controllers
    c. Models
    d. DTOs
    e. Services
    f. Data
    g. Program.cs
    h. appsettings.json
6. Actions
    a. Create Project
    b. Install Required Packages
    c. Create Model Classes
    d. You must use following Data Annotations wherever suits.
        i. Data Types
        ii. Nullable
        iii. Required
        iv. MaxLength

   v. EmailAddress
   vi. Phone
   vii. Range

e. Create DbContext: Inherit DbContext → Constructor → DbSets
f. Configure by overriding onConfiguring
g. Fluent API Configuration Inside OnModelCreating()
   i. Table Names
   ii. Primary Key
   iii. Required
   iv. Maximum Length
   v. Decimal Precision
   vi. Default Value
   vii. One To Many
      1. Customer → Policies
      2. PolicyType → Policies
      3. Policy → Claims
h. Seed by overriding OnModelCreating and modelBuilder.Entity<…>().HasData
   i. Policy Types: Health, Life, Vehicle, Travel, Home
   ii. 5 Customers
   iii. 6 Policies
   iv. 4 Claims
i. Create Controllers
   i. CustomersController
   ii. PoliciesController
   iii. PolicyTypesController
   iv. ClaimsController
j. Implement GET ALL, GET BY ID, POST, PUT and DELETE

ଓ✵⊞ Handle the data, route the request: turn tough code into your best.⊞✵ଓ